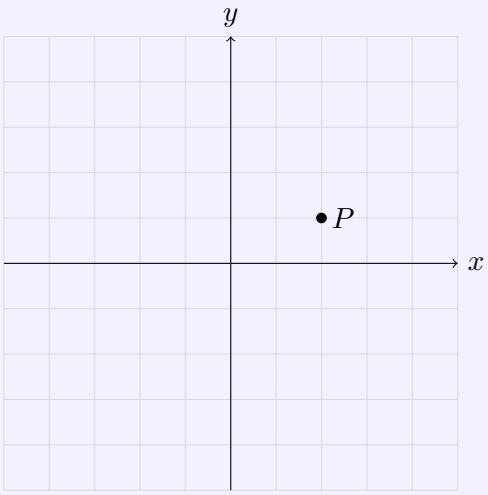
Algebra 1 – Transformations Worksheet

A. Warm Up – Directions from a Point

Find the new point after moving in the given direction.

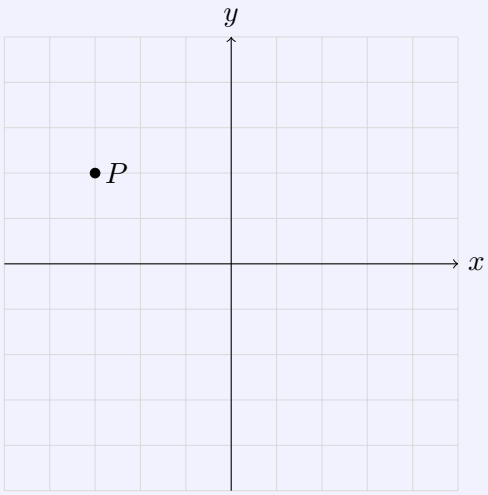
1. Point $P(2, 1)$

- Left 1: _____
- Right 2: _____
- Up 3: _____
- Down 4: _____



2. Point $P(-3, 2)$

- Left 1: _____
- Right 2: _____
- Up 3: _____
- Down 4: _____



Direction Rules

Left 1: $(x - 1, y)$

Right 2: $(x + 2, y)$

Up 3: $(x, y + 3)$

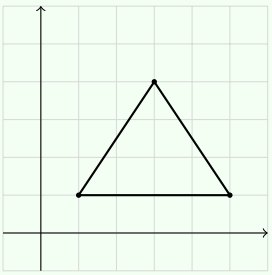
Down 4: $(x, y - 4)$

B. Stretch / Shrink (Dilations)

Graph the image after the transformation.

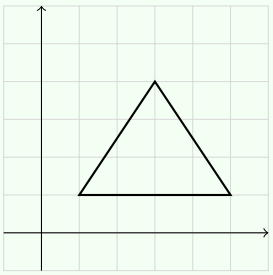
1. Horizontal Stretch by factor of 2

$(x, y) \to (2x, y)$



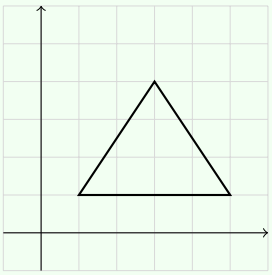
2. Horizontal Shrink by factor of $\frac{1}{2}$

$(x, y) \to \left(\frac{1}{2}x, y\right)$



3. Vertical Stretch by factor of 3

$(x, y) \to (x, 3y)$

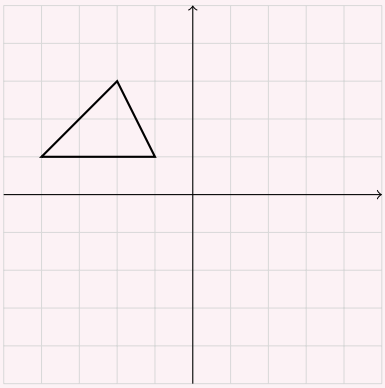


C. Reflections

Graph the reflected image.

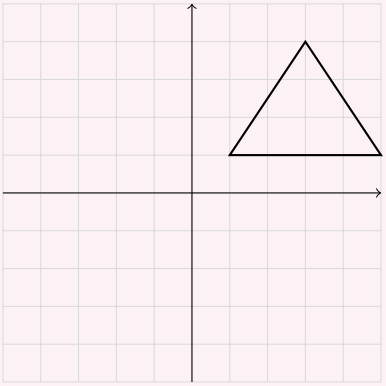
1. Reflect over the x-axis

$(x, y) \to (x, -y)$



2. Reflect over the y-axis

$(x, y) \to (-x, y)$



Reflection Rules

$$\text{Over } x\text{-axis} : (x, y) \to (x, -y)$$

$$\text{Over } y\text{-axis} : (x, y) \to (-x, y)$$

$$\text{Over } y = x : (x, y) \to (y, x)$$

$$\text{Over } y = -x : (x, y) \to (-y, -x)$$

Visual Summary

- **Translations** move a figure.
- **Dilations** change size.
- **Reflections** flip a figure.
- Shapes keep the same structure after transformations.

Practice Daily to Build Transformation Intuition!